

Nanyang Business School
BU5401:
Management Decision Tools

Seminar 13:
Decision Analysis

Decision Analysis

- Decision without Probabilities
- Decision with Probabilities – Decision Trees
- Bayes Theorem
- EVPI, EVSI

Decision Criteria

- Criteria
 - To maximize the maximum – optimistic
 - To maximize the minimum -- pessimistic,
 - To minimize regret,
 - To maximize/minimize the expected value, etc.
- Decision context
 - Decision situation --- the context affects the objectives and criteria
 - Decision maker

Payoff Table

Profit in \$millions

	S1	S2	S3
D1	0.5	0.4	0.3
D2	1.0	0.6	0.1
D3	2.0	0.5	-0.5

Optimistic and Pessimistic

Profit in \$millions

	S1	S2	S3	Max	Min
D1	0.5	0.4	0.3	0.5	0.3
D2	1.0	0.6	0.1	1.0	0.1
D3	2.0	0.5	-0.5	2.0	-0.5

Maximin Decision

Maximin Payoff

Maximax Decision

Maximax Payoff

Minimax Regret

Regret Table

	S1	S2	S3	Max
D1	1.5	0.2	0.0	1.5
D2	1.0	0.0	0.2	1.0
D3	0.0	0.1	0.8	0.8

Minimax
Regret
Decision

Minimax
Regret

Calculating Expected Values with a Payoff Table

Profit in \$millions

	S1	S2	S3	EV
D1	0.5	0.4	0.3	0.38
D2	1.0	0.6	0.1	0.49
D3	2.0	0.5	-0.5	0.35
Prob.	0.1	0.6	0.3	

Decision Tree

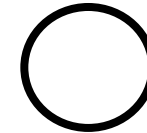
- A graphical representation of a decision problem according to the logical sequence in their time of occurrence for events
- Consists of Decision, Event, and Payoff nodes
- Drawn in a left-to-right manner indicating events unfolding from early to later stages.
- Able to display multi-stage decision problem by listing exhaustively all possible decision and event paths.

Notation and Symbol

Decision event: square



Chance (uncertain) event: circle



Value (payoff): rounded rectangle

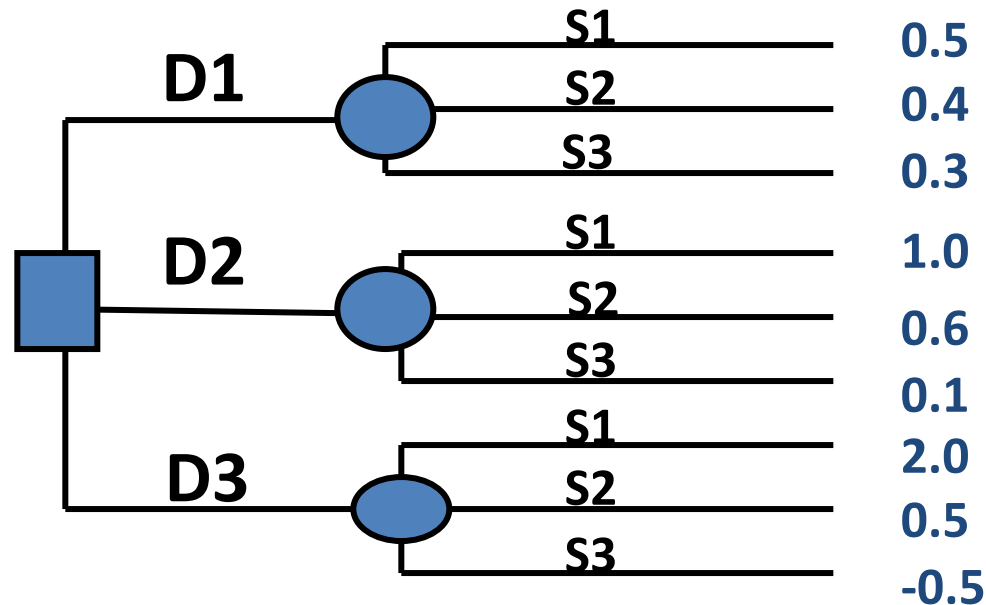


Link: pointed arrow or arc



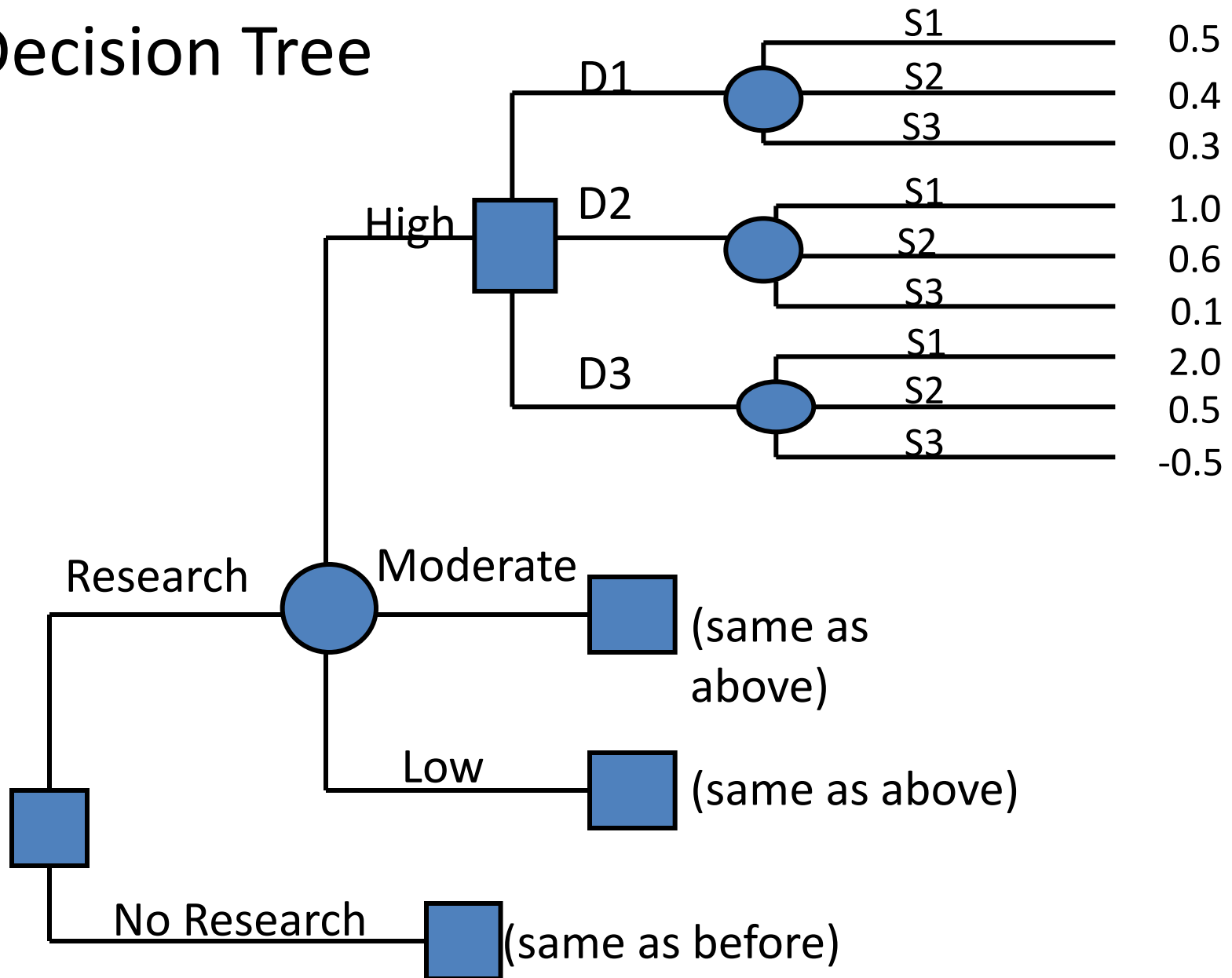
- Use a square for a decision event and a circle for a chance event
- Use a link leaving a chance event for a different outcome (state of nature), and a link leaving a decision node for a different decision alternative.
- Display the payoff at the end of each limb of a tree (the result for the series of branches that make up that limb).

Decision Tree: the Marketing Example



- To calculate correctly the expected value in order to make a correct decision at the root node, the payoff nodes should have net payoff after adding any bonus payoff and subtracting any cost payoff along the path.

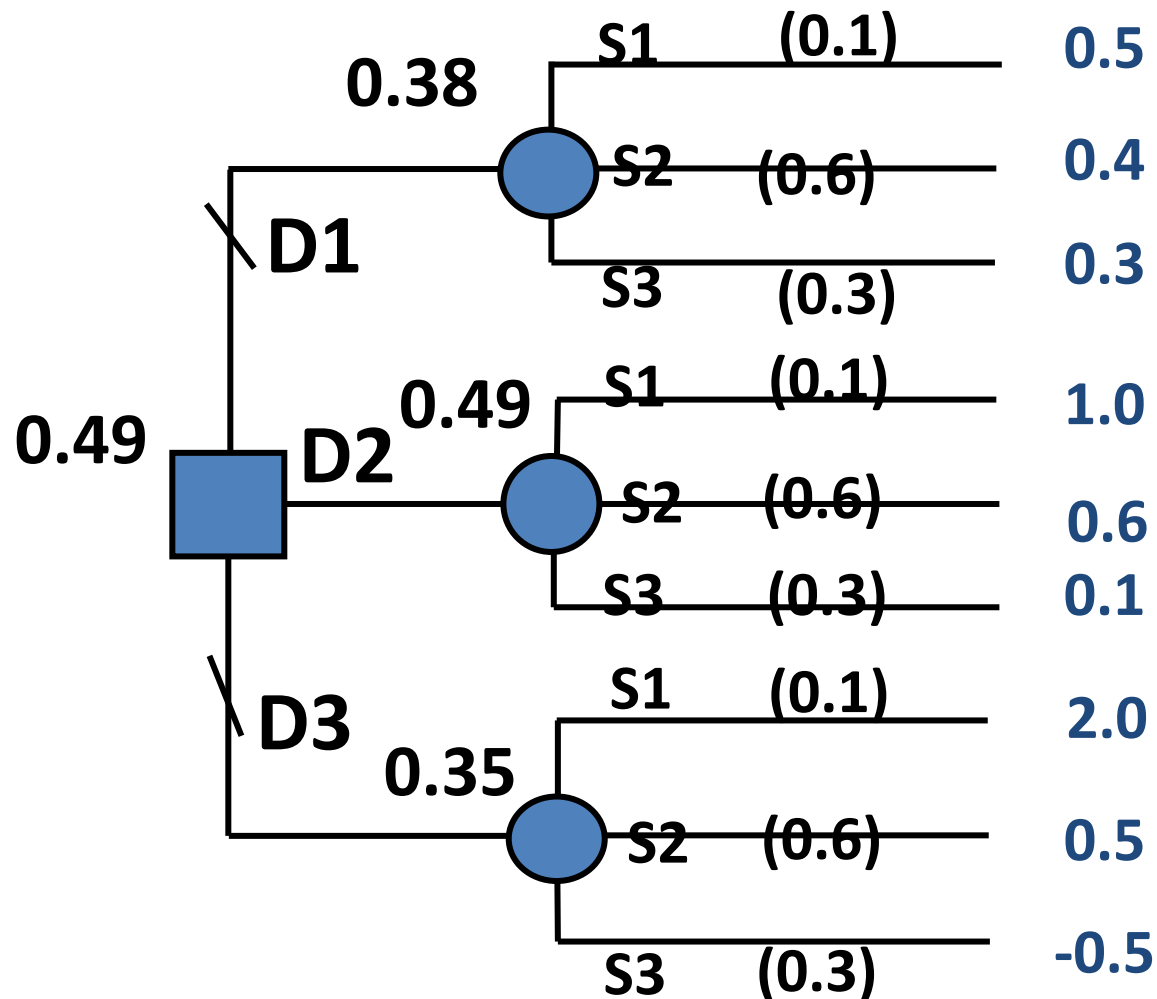
Decision Tree



Evaluating A Decision Tree

- Start from the “end branches” and work leftwards towards the “root”
 - For each chance node, calculate the EV and write the EV above the node
 - For each decision node, select and mark the decision with the maximum EV. Write the optimal EV above the decision node
- The optimal sequence of decisions is given by the route marked. The optimal EV is given above the “root” node

Marketing Example: Basic Case



Revision of Probabilities

Now suppose that you have hired a market research company to conduct a research. The research results indicate that the market acceptance level will be high. Will the probability for the market acceptance level to be actually high (or moderate, or low) with this forecast or will it still be the same as before?

The Bayes Rule

1. Prior belief
2. New information such as forecast, new research findings, experts' opinions, etc.
3. The prior belief is revised in the light of the new information
4. The updating depend on the prior probabilities and the reliability of the new information source

For the Marketing Example ...

1. Prior belief: $P(S1) = 0.1$, $P(S2) = 0.6$, $P(S3) = 0.3$
2. New forecast is high (denoted by H)
3. How reliable is the market research company?
4. How to revise the probabilities?

The Bayes Theorem

- S_j state of nature $j = 1, \dots, n$
- I_k market research indicator $k = 1, \dots, L$
- $P(I_k | S_j)$: reliability of the marketing research

$$P(S_j | I_k) = \frac{P(I_k | S_j)P(S_j)}{\sum_{t=1}^n P(I_k | S_t)P(S_t)}$$

Tableau method

- I_k , $k = 1, 2, 3$ the event that the market research company forecasts high, moderate, and low market acceptance level
- From past performance, the reliability of the market research company is given in the following table.
- Reliability is the conditional probability of the column event given the row event

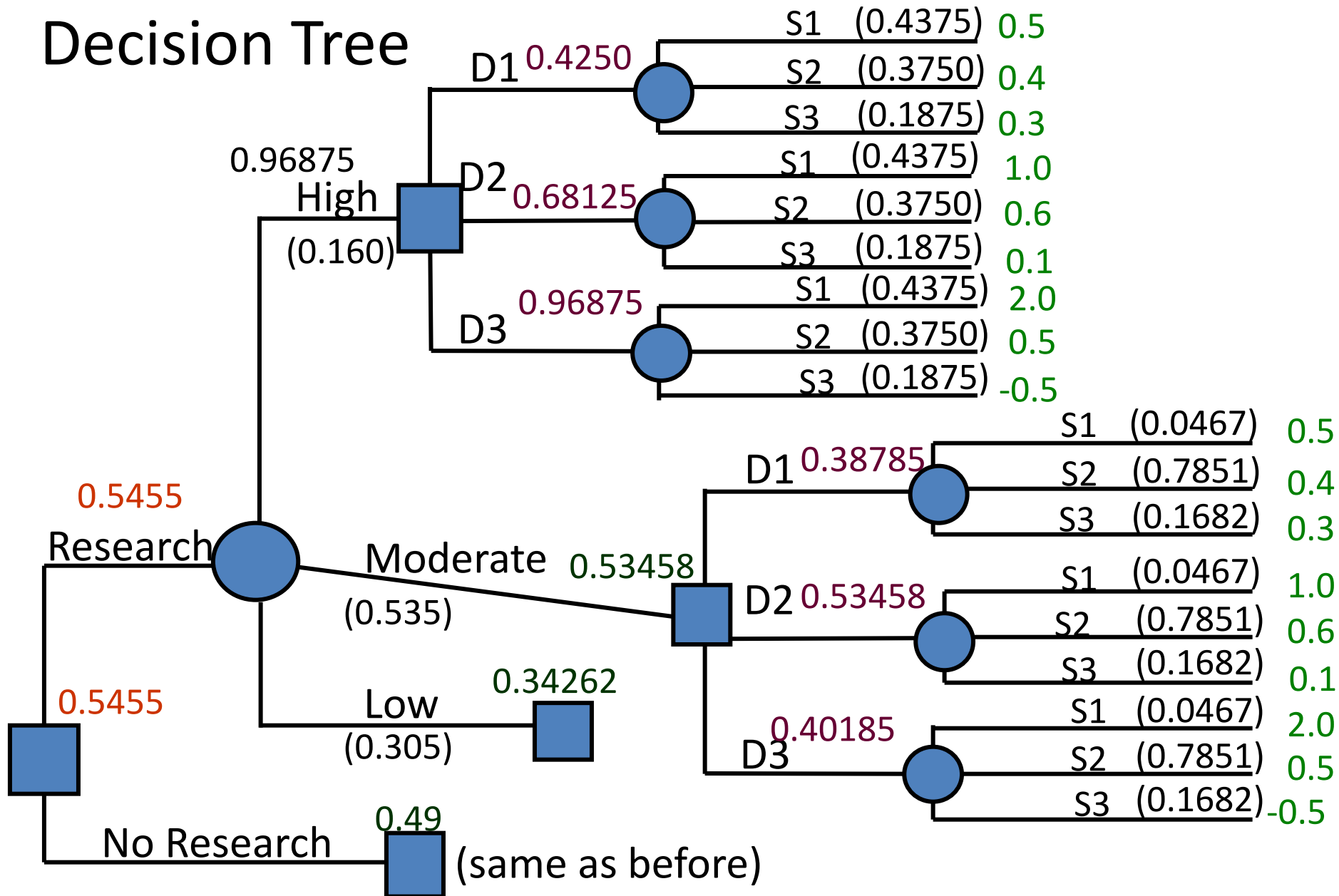
Reliability of the market research

	$P(I_k \mid S_j)$		
	I_1	I_2	I_3
S_1	0.70	0.25	0.05
S_2	0.10	0.70	0.20
S_3	0.10	0.30	0.60

Tableau

	$P(S_j)$	$P(I_k S_j)$			$P(I_k \cdot S_j)$			$P(S_j I_k)$		
		I_1	I_2	I_3	I_1	I_2	I_3	I_1	I_2	I_3
S_1	0.1	0.70	0.25	0.05	0.07	0.025	0.005	0.4375	0.0467	0.0164
S_2	0.6	0.10	0.70	0.20	0.06	0.420	0.120	0.3750	0.7851	0.3934
S_3	0.3	0.10	0.30	0.60	0.03	0.090	0.180	0.1875	0.1682	0.5902
$P(I_k)$					0.16	0.535	0.305			

Decision Tree



(same as above)

The Case of Perfect Information

Outcomes:

S1: High

S2: Moderate

S3: Low

Payoffs

S1 S2 S3

D1 0.5 0.4 0.3

D2 1.0 0.6 0.1

D3 2.0 0.5 -0.5

Prob. 0.1 0.6 0.3

D1: Singapore

D2: Asia

D3 International

0.2 0.36 0.09

EVwPI = 0.65

EVPI & EVSI

Expected Value of Perfect Information

- Expected Value with Perfect Information or
 $EVwPI = 2*0.1 + 0.6*0.6 + 0.3*0.3 = 0.65$
- Expected Value of Perfect Information or
 $EVPI = 0.65 - 0.49 = 0.16$

Had you known in advance that S1 is going to happen, what action will you take?

Expected Value of Sampled Information

- Expected value with sample information:
 $EVwSI = 0.5455$ (from the decision tree)
- Expected value of sample information:
 $EVSI = 0.5455 - 0.49 = \$0.0555 \text{ million} = \$55,000$
- Efficiency of Information
 $EVSI/EVPI = 0.0555/0.16 = 34.69\%$

Optimal strategy

- If the fee to be charged is greater than \$55,000 then no survey should be implemented and the optimal strategy is D_2 : market the product in Asia only
- If the fee to be charged is less than \$55,000 then the optimal strategy is to implement the survey. The optimal strategy is then dependent upon the sample outcome

What you have to write as answer

Survey outcome

- If I_1 : then D_3 market the product internationally
- If I_2 : then D_2 market the product in Asia
- If I_3 : then D_1 market the product in Singapore